

Qatar-1b

R-2611

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_190619 · created 2026-08-28T01:32:33+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458652.4022 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.056
Transit depth (Rp/Rs)^2	3.63 +/- 2.13 [%]
Orbital Inclination (inc)	88.71 +/- 2.87
Airmass coefficient 1 (a1)	1.078 +/- 0.061
Airmass coefficient 2 (a2)	-0.05 +/- 0.35
Scatter in the residuals of the lightcurve fit is	5.73 %
Best Comparison Star	#1 - [472, 453]
Optimal Aperture	24.76
Optimal Annulus	49.15
Transit Duration (day)	0.008 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.19 $\pm$ 0.056 vs 0.1463 (deviation 0.78 σ)
Duration fit vs published	0.008 d vs 0.0692 d (deviation 19.11 σ)
Mid-time O-C	-2048.5 min
Depth SNR	3.4
Residual scatter	5.73 %

Light curve

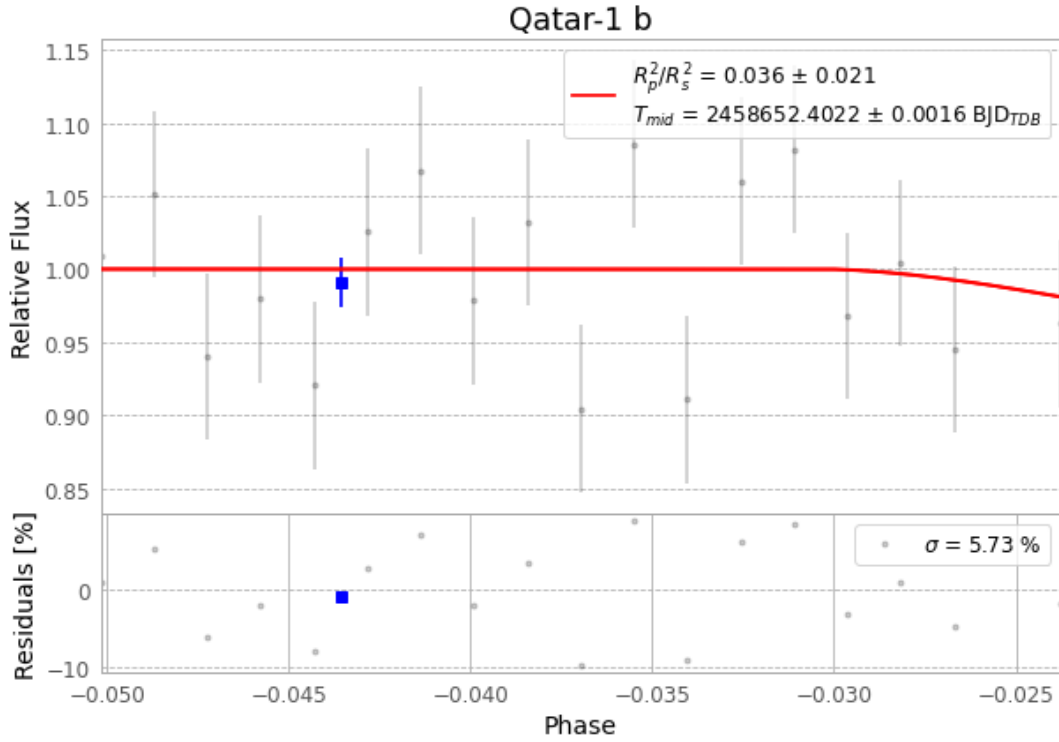


Figure 1 — FinalLightCurve_Qatar-1 b_2019-06-18.png (SHA-256 cd85a920c57734df...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 420], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 d886109f974038e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1190619060012.FITS → Qatar-1190619093612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190619.log (29f4d537850b...), FinalLightCurve_Qatar-1 b_2019-06-18.png (cd85a920c577...), FinalLightCurve_Qatar-1 b_2019-06-18.csv (28f2c2d1f071...)

Compute resources

Wall time 225.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 01:32Z.